

Do the numbers mean what you think they mean?

Francisco Villamil

Research Design for Social Sciences
MA Computational Social Science, UC3M
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Where we are

1. Why design at all?
2. What is the question, and where would the data come from?
3. **Do the numbers mean what you think they mean?**
4. What comparisons can we use to establish causal claims?
5. What variation can you exploit, and how far does the answer travel?

Where we left it

Google Flu Trends counted searches.

Everyone talked about it as if it counted flu.

- The gap between *what you count* and *what you mean* is today's topic
- No amount of data closes it. Only design does

Roadmap

Concepts and operationalization

Measuring the right stuff

Latent variables and model outputs

Who is missing, and at which level

Description as a goal

Synthesis: wartime civilian deaths

Paper: Roads to Rule, Roads to Rebel

Before next week

Concepts

- Concepts are the **building blocks** of arguments
 - part of the theory, not of the data
- Sometimes easy (income), usually not
 - *Household* income? What counts as a household?
 - Political violence? Ideology? Democracy? Polarization?
- Concept $\neq$ term
 - “authoritarian regime”, “dictatorship”, “autocracy”: same concept?

Two ways of thinking about a concept

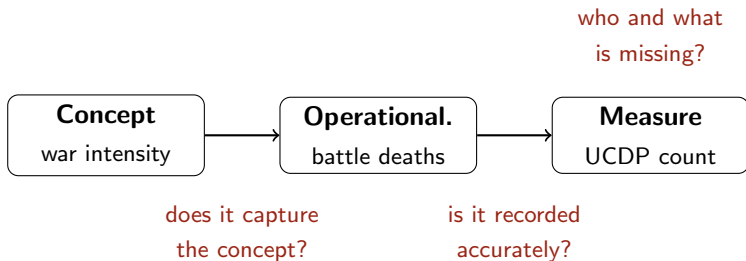
1. **Rule-based:** a definition, with necessary conditions
 - what are the rules that make something a household?
 2. **Ideal type:** what does a typical case look like?
 - can we recognise a household intuitively, and say how close a case is?
- Not two kinds of concepts, but two tools to sharpen one

Try it: **political violence** or **terrorisk attack**

Operationalization

- Translating an abstract concept into something **observable**
 - concept: war intensity → operationalization: battle deaths
- Operationalize $\neq$ measure
 - knowing *what* you would observe does not mean you *can* observe it
 - ideology of a Twitter user: easy to operationalize, hard to measure

Three steps, three places to fail



Deaths from famine and disease? Civilians? Wars below 25 deaths/year?

Your turn

- Your question is about some cause of **civil war outbreak**
- Two concepts hiding in there:
 1. **Civil war**
 2. **Outbreak**
- How would you define each? How would you operationalize it?

Why this is not a philosophy detour

- A large share of good quantitative work *is* better concepts and better measures
- Especially in CSS: new data mostly means **new ways to measure old concepts**
 - ideology from follow networks, wealth from phone records, state reach from road maps
- Today's paper is exactly this

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Three measurement questions

1. Are you measuring **what you think** you are measuring?
 - proxies, latent variables, model outputs
2. Who or what are you **not** measuring?
 - missing data, selection
3. At **which level**?
 - the unit of analysis, again

How was this number made?

- We usually look at a variable without asking how it was created
- Look closer:
 - survey wording
 - coding rules (who decided what counts as a democracy?)
 - raw material (news reports? experts? admin records? posts?)
- Above all: are there **biases related to your question?**

Example: measuring discrimination

- Question: does discrimination of minorities hurt a country's economic performance?
- You find a dataset scoring each minority's discrimination, 0 to 10
 - codebook: “unequal access to state power, from active discrimination (incl. mass violence) to lack of access to key positions”
- Coded by sending a questionnaire to 2–3 country experts
- **Thoughts?**

Same data, different question

- Now: does **extreme** discrimination make violence against minorities more likely?
 - violence from another dataset, coded from newspapers
- You find a **positive relationship**
- Go back to the definition
- And: comparing *within* countries, or *between* them?

Has democracy declined?

- Widespread claim: the world is going through **democratic backsliding**
- Problem: how do we measure democracy?
- Most global indices rely on **expert judgement**

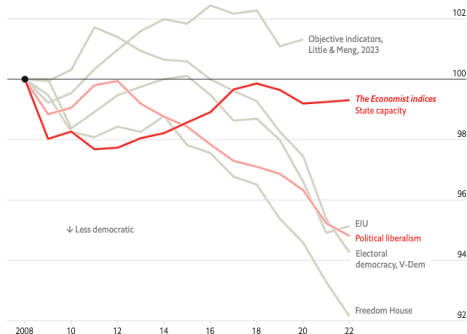
Graphic detail | Measure for measure

Global democratic backsliding seems real, even if it is hard to measure

Our analysis highlights two measures of governance that have diverged in recent years

Sep 12th 2023

Global democracy indices, 2008=100



Measuring Democratic Backsliding

Andrew T. Little, *University of California, Berkeley, USA*

Anne Meng, *University of Virginia, USA*

ABSTRACT

Despite the general narrative that the world is in a period of democratic decline, there have been surprisingly few empirical studies that assess whether this is systematically true. Most existing studies of global backsliding are based largely if not entirely on subjective indicators that rely on expert coder judgment. Our study surveys objective indicators of democracy (e.g., incumbent performance in elections) and finds little evidence of global democratic decline during the past decade. To explain the discrepancy in trends between expert-coded and objective indicators, we consider the role of coder bias and leaders strategically using more subtle undemocratic action. Although we cannot rule out the possibility that the world is becoming less democratic exclusively in ways that require subjective judgment to detect, this claim is not justified by existing evidence.

Two operationalizations of democracy

‘Objective’

Minimalist concept: contested elections

e.g. Democracy & Dictatorship:

1. elected executive
2. elected legislature
3. more than one party
4. alternation in power has happened

But: incumbents losing elections is not so objective either (young, growing democracies)

‘Subjective’

Maximalist concept: elections, plus rule of law, media, participation. . .

e.g. V-Dem: expert surveys, several coders per country, with measures of uncertainty

But: autocrats are creative, and experts read the same newspapers

Proxies

- A **proxy** stands in for something you cannot observe
- Creative, useful, and always a bet. What could bias these?
 - Economic development $\leftarrow$ night-time lights from satellites
 - Household wealth $\leftarrow$ owning a TV
 - Emigration (early 20th c. Spain) $\leftarrow$ missing men in the census

A proxy that encoded the wrong thing

- US hospitals used an algorithm to decide who gets **extra care**
→ algorithms like this reach ~ 200 million people a year in the US
- Target: patients with the greatest **health need**
- What it actually used as a proxy: **healthcare cost**
- (The model was set to predict future healthcare costs, and so its finding could be interpreted to foresee healthcare needs)
- What could go wrong?

Obermeyer, Powers, Vogeli & Mullainathan (2019), *Science* 366(6464).

Obermeyer et al.: what went wrong

- People with better resources could spend more even if they had the exact same illness
- They did not include race in the models, deliberately so as not to discriminate
- But at equal illness, **less was spent** on Black patients
 - unequal access to care, not less need
- So at the same risk score, Black patients were **considerably sicker**
- Measuring need directly would have raised the share of Black patients flagged from **18% to 47%**

The proxy was accurate. It was accurate about the wrong thing.

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Latent variables

- Some concepts cannot be observed directly, at all
 - ideology, trust, state capacity, polarization
- Instead: **infer** them from several things you *can* observe (i.e. from their traces)
- Compare classic strategies for, e.g., party positions:
 - expert surveys
 - coding texts (manifestos, speeches)
 - roll-call votes (**latent variable approach**)
- Each captures a slightly different concept

Example: ideology on Twitter

- Do left- and right-wing users behave differently online?
- The outcome is easy: content, frequency, networks
- **But how do you measure ideology?**
 - only politicians, where you know it? (selection)
 - link survey responses to accounts, with consent? (cost, non-response)
 - **infer it from behaviour?** (example from Barberá 2015, *Pol Analysis*)

2 Ideal Point Estimation Using Twitter Data

2.1 *Assumptions*

In this article, I demonstrate that valid ideal point estimates of individual Twitter users and political actors with a Twitter account can be derived from the structure of the “following” links between these two sets of users. In order to do so, I develop a Bayesian spatial model of Twitter users’ following behavior.

The key assumption of this model is that Twitter users prefer to follow politicians whose positions on the latent ideological dimension are similar to theirs. This assumption is equivalent to that of spatial voting models (see, e.g., Enelow and Hinich 1984). I consider following decisions to be costly signals about users’ perceptions of both their ideological location and that of political accounts. Such cost can take two forms. If the content of the messages users are exposed to as a

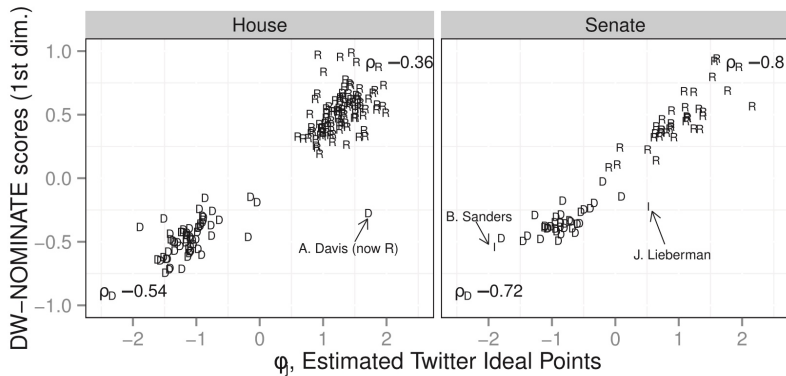


Fig. 1 Ideal point estimates for members of US Congress.

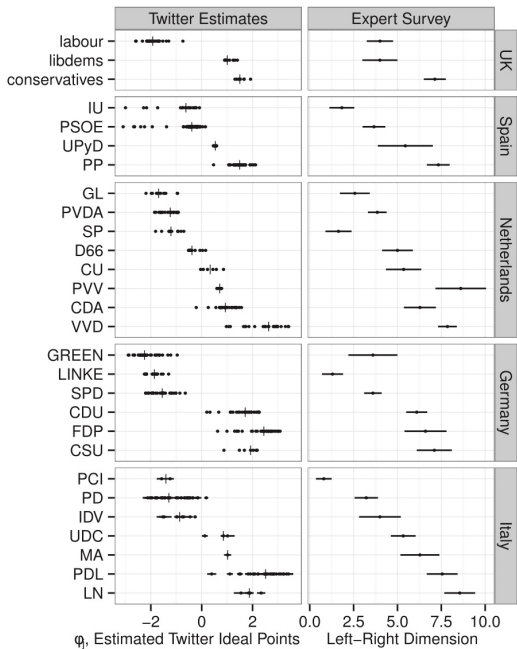


Fig. 3 Ideological location of parties in five European countries.

When your variable is a model's output

- Increasingly common: a variable that is the **prediction of a model**
 - sentiment, toxicity, topic, stance, ideology
 - inferred gender, age, location
 - labels produced by an LLM
- The measurement question does not go away. It gets **harder**
- Because the error is almost never random

Non-random errors in CSS

- A toxicity classifier flags more posts as toxic when written in some dialects
 - “group X is more toxic” is partly “the classifier reads group X badly”
- A sentiment model trained on English, applied to Spanish and Catalan
 - the comparison across languages is partly a comparison of accuracy
- An LLM labels posts, then the model is updated
 - drift, last week: the instrument changed underneath you
- See Knox et al (2022), ‘Testing Causal Theories with Learned Proxies’ for more

The rule: validate, always

- There is no automatic method that is valid everywhere
 - the “no free lunch”
- Validate against something **external**:
 - a hand-coded sample (by humans, blind to the model)
 - a known group (politicians, registered party members)
 - another measure built from different data
- And validate **in the population you will use it on**

Grimmer, Roberts & Stewart (2022), *Text as Data*.

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Missing data

- Almost always more important than it looks
1. Missing **completely at random** (MCAR)
 - no pattern at all. Rare
 2. Missing **at random** (MAR)
 - depends only on things you *observe*, so you can adjust for it
 3. Missing **not at random** (MNAR)
 - missingness is related to the thing you are studying

Missing data: example

- **Our question:** Does some habit reduce post-surgery recovery time?
- **How we get the data:** Forms filled in at hospital check-in, some are missing. Why?
 1. A random batch of forms got lost $\Rightarrow$ MCAR (although...)
 2. One (recorded) admin worker always forgets to ask $\Rightarrow$ MAR (although...)
 3. Emergencies do not fill in forms $\Rightarrow$ MNAR

Selection, again

- Sometimes the sample was designed with a bias
 - an online survey and people over 65
- Harder to see: an **invisible filter** decides who is in the data
 - who consents to share their feed with researchers
 - who posts at all
- Remember the Xbox poll: the skew is fine **only if you know it and can correct for it**

At which level?

- The unit of analysis, again
- Not all variables live at the same level
 - individual data, household income; tweet data, user ideology
- Choose the level that matches the **theory and mechanism**
 - e.g. wealth and voting in the US, state-level vs individual-level
- Measuring at the wrong level can change the answer

First half in four lines

- Concept $\rightarrow$ operationalization $\rightarrow$ measure
- Error is fine. **Error related to your question** is not
- Model outputs are measures. Validate them
- Ask who is missing, and at which level

Questions?

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Is description useful?

- Not every question needs a *why*
 - has political violence increased? how segregated are friendships?
how many people are 3 steps away?
- Description is not the consolation prize for failing to find a cause
- It has its own design requirements:
 - a clear **population**
 - a defensible **denominator**
 - **coverage** and **comparability** across units and time

Before anything: look at the data

- These are **real-world observations**. Plot your main variables
- Is the distribution what the theory expects?
 - bimodal where you expected one peak?
 - a spike at 0, or at 99?
 - a jump in one year?
- The shape can say something about the **mechanism**
 - the effect of income in a very unequal vs. a very equal society

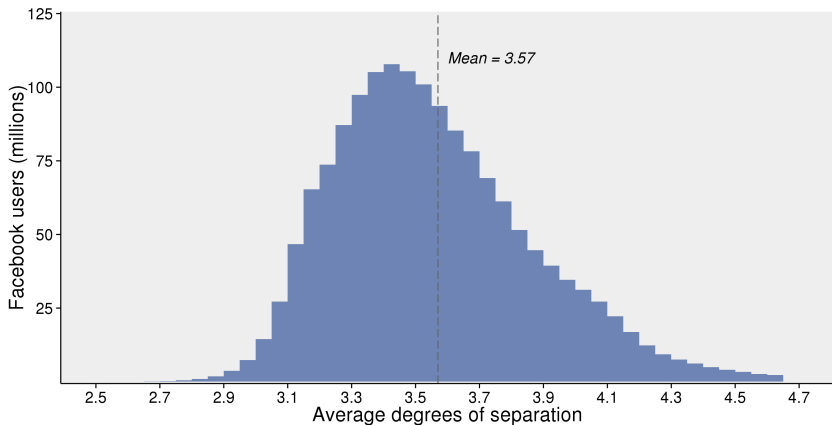
A descriptive theory

I read somewhere that everybody on this planet is separated by only six other people. Six degrees of separation. Between us and everybody else on this planet. The president of the United States. A gondolier in Venice.

John Guare, *Six Degrees of Separation*

- A claim with no *why* in it. Can we check it?

Facebook: 3.57 degrees



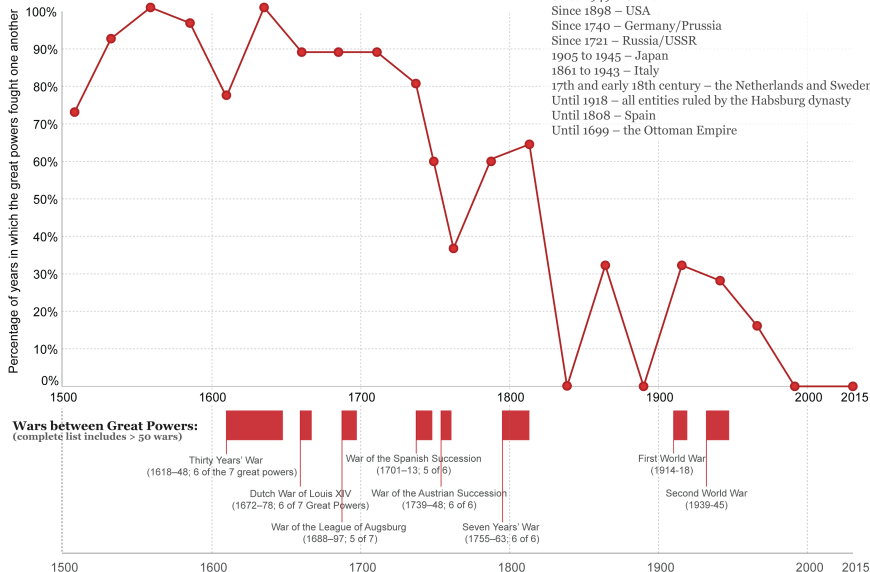
Average degrees of separation between Facebook users, 2016.

Percentage of years in which the 'Great Powers' fought one another, 1500-2015 – by Max Roser

Between 1500 and today there were more than 50 wars between 'Great Powers'.
Data are aggregated over 25-year periods.

The Great Powers:

- Entire period – France and England/Great Britain/U.K.
- Since 1949 – China
- Since 1898 – USA
- Since 1740 – Germany/Prussia
- Since 1721 – Russia/USSR
- 1905 to 1945 – Japan
- 1861 to 1943 – Italy
- 17th and early 18th century – the Netherlands and Sweden
- Until 1918 – all entities ruled by the Habsburg dynasty
- Until 1808 – Spain
- Until 1699 – the Ottoman Empire



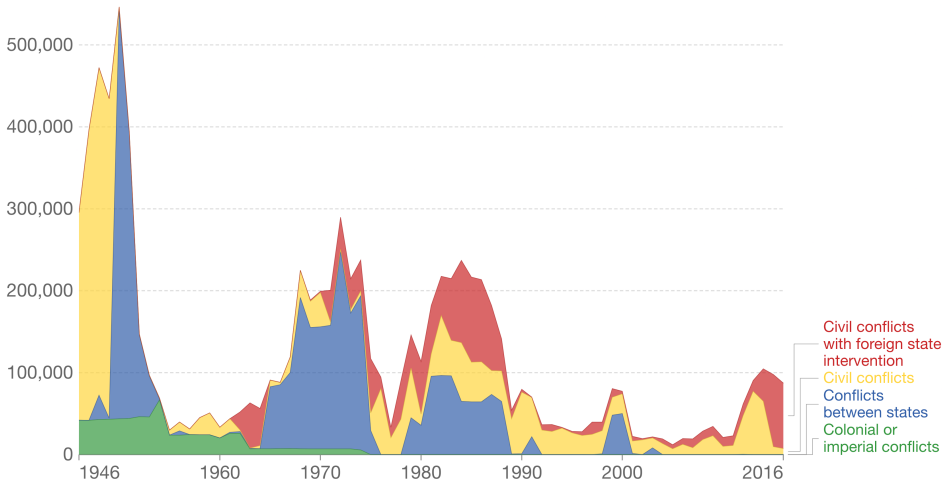
Data source: Steven Pinker (2011) – The Better Angels of Our Nature: Why Violence Has Declined. Based on data from Levy, J. S., & Thompson, W. R. (2011) – The Arc of War

The interactive data visualisation is available at OurWorldinData.org. There you find the raw data and more visualisations on this topic.

Licensed under CC-BY-SA by the author Max Roser.

Battle-related deaths in state-based conflicts since 1946, 1946 to 2016

Only conflicts in which at least one party was the government of a state and which generated more than 25 battle-related deaths are included. The data refer to direct violent deaths. Deaths due to disease or famine caused by conflict are excluded. Extra-judicial killings in custody are also excluded.



Source: UCDP/PRIO

CC BY

Note: The war categories paraphrase UCDP/PRIO's technical definitions of 'Extrasystemic', 'Internal', 'Internationalised internal' and 'Interstate' respectively. In a small number of cases where wars were ascribed more than one type, deaths have been apportioned evenly to each type.

Description at scale: social capital

- 21 billion Facebook friendships, 72 million US users aged 25–44
- Concept: **economic connectedness**
 - share of friends with high socioeconomic status among people with low SES
- Mapped by ZIP code, county, high school, college
- Among the strongest predictors of upward income mobility they found

Chetty et al. (2022), Social capital I, *Nature* 608.

The design decisions behind the map

- **Population:** Facebook users aged 25–44. Who is not there?
- **SES:** not observed. **Predicted** from many signals by a model
 - a model output, used as a variable. Validated against external data
- **Denominator:** share of friends, not number of friends
- **Access:** nobody else can rebuild it from the raw data
 - released only as aggregates, with noise added for privacy

Culturomics

- Google Books: millions of digitised books, word frequencies by year
- “Quantitative analysis of culture”: fame, censorship, the spread of ideas
- e.g. the use of scientific terms rises steadily through the 20th century
- Is that a fact about **culture**, or about **the corpus**?

Michel et al. (2011), *Science* 331.

What was actually measured

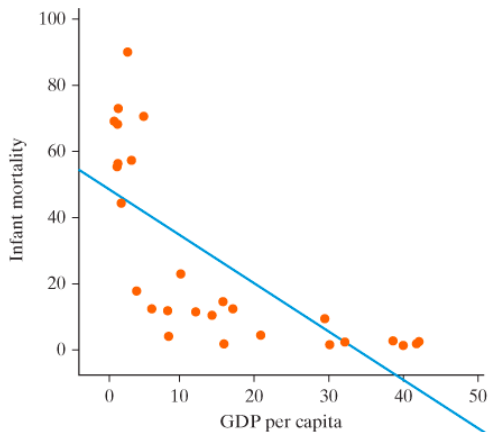
- The corpus is a **library**, not a record of what people read
 - one copy per book, whether it sold ten copies or ten million
- Its composition **changes over time**
 - scientific publishing takes up a growing share through the 20th century
- So a trend in words is partly a trend in **what got published and digitised**

Pechenick, Danforth & Dodds (2015), *PLoS ONE* 10(10).

Describing relationships

- A relationship: knowing one variable tells you something about another
- A friend is bringing two kids in your small car. You ask:
 - “How old are they?” “6 and 2.” Now you know something about size
 - “What colour is their hair?” You learn nothing about size
- That is all a statistical relationship is: **conditional** values

Describing relationships



Infant mortality at \$1,000 GDP per capita? At \$30,000?

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World

Africa

Americas

Asia

Australia

China

Europe

India

Middle East

More ▾

Ukraine says it has identified a Russian commander accused of Bucha atrocities



By Mariya Knight and Radina Gigova, CNN

🕒 2 minute read · Published 1:04 AM EDT, Sat September 2, 2023



Groups of four

- Intuition: armed groups treat civilians better when they **need their resources** (labour, food, information) to survive
1. Clean up the theory. What are the main **concepts**?
 2. How would you **operationalize** them? What would you measure?
 3. At which **level**? Who would be **missing**?
 4. What **descriptive** finding would already be worth knowing?

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Roads to Rule, Roads to Rebel: Relational State Capacity and Conflict in Africa

Carl Müller-Crepon¹ , Philipp Hunziker²,
and Lars-Erik Cederman³

Abstract

Weak state capacity is one of the most important explanations of civil conflict. Yet, current conceptualizations of state capacity typically focus only on the state while ignoring the relational nature of armed conflict. We argue that opportunities for conflict arise where relational state capacity is low, that is, where the state has less control over its subjects than its potential challengers. This occurs in ethnic groups that are poorly accessible from the state capital, but are internally highly interconnected. To test this argument, we digitize detailed African road maps and convert them into a road atlas akin to Google Maps. We measure the accessibility and internal connectedness of groups via travel times obtained from this atlas and simulate road networks for an instrumental variable design. Our findings suggest that low relational state capacity increases the risk of armed conflict in Africa.

In pairs/groups

1. What is the concept, in one sentence, **without using the word “roads”**?
2. What does the road network **fail to capture**?
3. Tell a story where their measure **goes up** and actual state capacity **goes down**

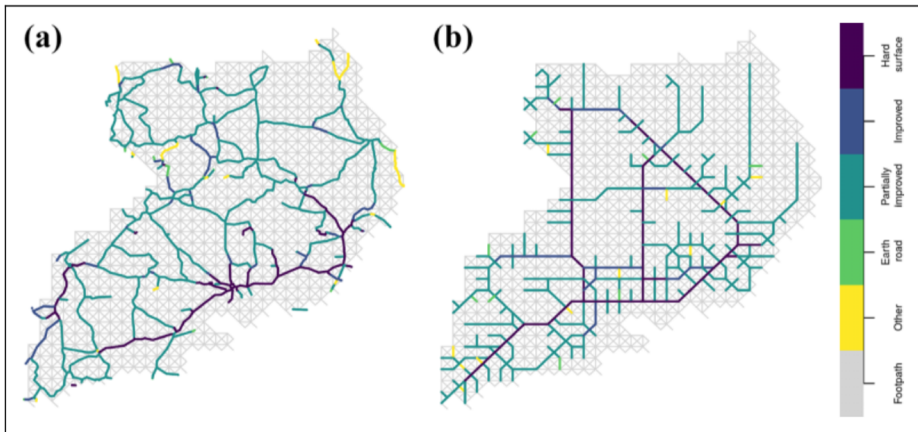


Figure 3. Observed and simulated road network in Uganda, 1966. (a) Observed network. (b) Simulated network.

Groups of four: build it today

You want to measure relational state capacity
in 2026, not in 1966.

What data would you use, and what would it miss?

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The three goals, after today

- **Describe:** measurement *is* the design
- **Explain:** a biased measure can manufacture an effect
- **Predict:** an accurate prediction of the wrong thing is still wrong

Next week

We can describe a relationship.

Next week: when does it tell us that one thing **causes** another?

Before next week

- **Read:** Guess *et al.* (2023), *How do social media feed algorithms affect attitudes and behavior in an election campaign?*
 - Skip the statistics. Read for **what was randomized**, and **who took part**
- **Memo 3**, in pairs, Aula Global by **Monday night**:
 - Pick one concept (not democracy)
 - Operationalize it two different ways, with data that exists
 - Which would you choose, and what do you lose?
 - Half a page

Questions?